

Explainable AI-Generated Music Detection Using MERT and Attention Pooling

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Abstract

Generative artificial intelligence systems such as Suno and Udio can now synthesise complete songs that are difficult to distinguish from human-produced music, creating acute risks for copyright enforcement and artist livelihoods. Existing AI-generated music detectors suffer from poor cross-generator generalisation, fragility under routine audio transformations, and a lack of interpretability. This work reviews twenty recent peer-reviewed studies spanning music foundation models, AI-music and synthetic-song detection, singing-voice deepfake detection, self-supervised audio anti-spoofing, and explainability methods for audio classifiers, in order to motivate a proposed detector built on the MERT self-supervised music foundation model with attention pooling and a three-level explainable AI (XAI) layer combining attention weights, Grad-CAM spectrogram localisation, and SHAP-based layer analysis.

Keywords: AI-generated music detection, explainable AI, MERT, attention pooling, audio deepfake detection, self-supervised learning

1. Introduction to the Research Problem

Generative artificial intelligence has reached the point where tools such as Suno and Udio can synthesise complete songs – vocals, instrumentation, lyrics, and stylistic arrangement – that are difficult for human listeners to distinguish from professionally produced human music. While these systems democratise music creation, their unregulated use poses acute risks to the music industry, to copyright enforcement, and to the economic livelihood of human artists. The scale is no longer hypothetical: streaming platforms now

report that a substantial fraction of newly uploaded tracks are AI-generated, with millions of such tracks identified and tagged across a single year. This has converted AI-generated music detection from an academic curiosity into an operational requirement for distributors, rights organisations, and digital service providers.

Despite rapid progress, the current generation of detectors suffers from three interlocking weaknesses. First, generalisation across generators is poor: a detector trained on one generative family typically collapses when evaluated on songs from a different family, with reported F1 scores falling from roughly 0.99 in-distribution to about 0.629 cross-generator. Second, robustness to real-world transformations is fragile: routine operations such as MP3 compression or pitch shifting can severely degrade detector accuracy, even though such transformations are ubiquitous in distribution pipelines. Third, and most consequentially for deployment, existing detectors are opaque: they provide no human-interpretable justification for why a track was flagged as synthetic, which makes them untrustworthy in settings where a false accusation can damage an artist’s reputation or withhold their income.

This survey situates a proposed solution to these weaknesses within the existing literature. The proposed approach builds a detector on top of MERT, a self-supervised music foundation model, using attention pooling to aggregate frame-level representations, and augments the classifier with a three-level Explainable AI (XAI) layer – combining attention weights, Grad-CAM spectrogram localisation, and SHAP-based layer analysis – so that every decision is transparent and auditable. To motivate that design, the remainder of this document reviews twenty recent, peer-reviewed studies spanning music foundation models, AI-music and synthetic-song detection, singing-voice deepfake detection, self-supervised audio anti-spoofing, and explainability methods for audio classifiers.

2. Literature Review

The reviewed literature clusters into five themes that together define the technical landscape for explainable AI-music detection: (i) music representation and foundation models, (ii) AI-generated music and synthetic-song detection, (iii) singing-voice deepfake detection, (iv) self-supervised representations and pooling for audio anti-spoofing, and (v) explainability and robustness for audio detectors.

2.1. Music Representation and Foundation Models

MERT [13] is the cornerstone of the representation layer. It introduces a BERT-style transformer trained with a multi-task masked-prediction objective, guided by a Residual Vector Quantisation Variational Autoencoder (RVQ-VAE) acoustic teacher and a Constant-Q Transform (CQT) musical teacher that injects pitch and harmonic inductive bias. MERT achieves state-of-the-art or competitive results across fourteen music information retrieval (MIR) tasks while remaining computationally affordable, and its 95M-parameter public variant performs comparably to far larger models. Crucially for this project, MERT is released as a reusable feature extractor rather than a detector, which makes it a natural frozen backbone over which a detection head and explainability layer can be built. Complementary layer-wise analyses of self-supervised music models confirm that different transformer layers encode a transition from low-level acoustic to higher-level semantic features, motivating learned aggregation across layers rather than naive use of a single layer.

2.2. AI-Generated Music and Synthetic-Song Detection

The base paper, the AI Music Arms Race [7], provides the central problem statement and the quantitative target. It benchmarks detector architectures under an explicit cross-generator protocol and demonstrates the dramatic in-distribution-to-cross-generator collapse (F1 0.99 to 0.629) that this project seeks to narrow. SONICS [14] supplies the primary dataset and a complementary architectural insight: it assembles over 97,000 songs (4,751 hours) of real and end-to-end AI-generated music from Suno and Udio, and introduces SpecTTTra, a spectro-temporal tokeniser transformer that improves long-range temporal modelling, outperforming a ViT baseline by 8% F1 on long songs while reducing memory and latency. Afchar et al. [2] reframe detection as identifying the fingerprint of a neural decoder, training on real-versus-autoencoded-reconstruction pairs to isolate codec artefacts from musical content; they reach 99.8% accuracy yet candidly show that this accuracy is fragile under distribution shift – the “easy but actually hard” phenomenon. FakeMusicCaps [6] extends the problem to detection and source attribution across multiple text-to-music generators, reporting strong closed-set performance but marked degradation in open-set (unseen-generator) conditions. Two further works from the Schuller group bracket the field: a pathway survey [11] that argues for transferring audio-deepfake foundation models to music and for relying on intrinsic musical features over watermarking, and

an explainable-detection benchmark [12] that is among the first to evaluate explanation quality alongside detection accuracy for machine-generated music. A frequency-domain study, the Fourier explanation of AI-music artefacts [1], grounds detectability in interpretable spectral evidence, providing a principled account of the cues a detector should rely upon.

2.3. Singing-Voice Deepfake Detection

Singing-voice deepfake detection (SVDD) is the closest predecessor task and supplies hard lessons about domain shift. SingFake [19] defines the SVDD task and releases an in-the-wild dataset of authentic and deepfake song clips across multiple singers and languages; it shows that speech-trained anti-spoofing models degrade sharply on sung audio, especially under unseen singers, communication codecs, and background accompaniment. CtrSVDD [18] complements this with a controlled benchmark of clean, unaccompanied vocals that isolates true synthesis artefacts from voice-separation artefacts, and underpins the inaugural SVDD Challenge. Together these works demonstrate that authenticity detectors must be evaluated across conditions and that artefacts introduced by pre-processing can confound naive classifiers – a warning directly relevant to robust music detection.

2.4. Self-Supervised Representations and Pooling for Audio Anti-Spoofing

A mature line of speech anti-spoofing research establishes the value of self-supervised front-ends and of learned pooling – the two pillars of the proposed method. Zhang et al. [20] pair an XLS-R front-end with a Sensitive-Layer-Selection classifier that exploits multi-layer features, achieving state-of-the-art equal error rate on the ASVspoof 2021 deepfake set and demonstrating that intelligent use of intermediate layers outperforms last-layer features. Guo et al. [10] combine WavLM with a multi-fusion attentive classifier, providing direct evidence that attention-based pooling improves cross-condition robustness over simple temporal averaging – precisely the inductive choice this project adopts for aggregating MERT frames. El Kheir et al. [8] add a comprehensive layer-wise analysis across self-supervised models that quantifies which layers carry the most discriminative information, while Xie et al. [16] attack the generalisation problem head-on with a domain-generalisation framework that aggregates genuine features and separates spoofing artefacts to improve unseen-attack performance. Robustness to transmission conditions is examined by Shi et al. [15], who benchmark detector degradation

under real-world communication codecs and channels, reinforcing the need to evaluate under compression rather than only on clean audio.

2.5. *Explainability for Audio Detection*

The explainability cluster supplies the methods for the proposed three-level XAI layer. Channing et al. [5] introduce an attention-rollout mechanism for transformer audio classifiers together with a cross-corpus generalisation benchmark, explicitly framing transparency as a prerequisite for real-world trust. Grinberg et al. [9] develop time-domain relevance attribution that reveals which temporal segments drive a detector’s decision, benchmarking it against Grad-CAM, DeepSHAP, and GradientSHAP baselines. Yu et al. [17] apply SHAP to spectrogram- and waveform-trained networks to expose the acoustic features behind spoof decisions, and Bisogni et al. [4] build explainable detectors on interpretable acoustic features with explicit feature-importance analysis. Finally, Bajwa et al. [3] combine a Mel-spectrogram CNN with Grad-CAM-style visual explanation to localise fake-indicative time-frequency regions. Collectively these studies validate the three explanation modalities – attention weights, Grad-CAM spectrograms, and SHAP layer analysis – that the proposed system integrates, while also revealing that explanation fidelity remains an open metric and that almost all of this work targets speech rather than full-length music.

3. Comparative Analysis of Existing Methods

Table 1 contrasts ten representative studies across approach, dataset and domain, and headline result or limitation. The base paper is highlighted. The comparison exposes a clear pattern: methods that achieve the strongest raw accuracy (Afchar et al.; SONICS) do so within a single generator family or under limited transformation, whereas methods that emphasise generalisation or explainability (Channing et al.; Xie et al.; the XAI studies) originate largely in the speech domain and have not been validated on full-length music built on a music foundation model.

Three observations follow from this comparison. First, the representation layer and the detection objective are usually decoupled: MERT delivers powerful music features but is not a detector, while the strongest detectors use bespoke spectrogram or codec-fingerprint pipelines rather than a music foundation model. This leaves an obvious opening to combine MERT features

Table 1: Comparative analysis of representative studies. The base paper is marked with an asterisk (*).

Sl.	Study (Year)	Approach / Methods	Dataset & Domain	Key Result / Limitation
1	MERT [13]	SSL transformer; RVQ-VAE + CQT teachers; masked prediction	Music4ALL / large music corpora; MIR	SOTA on 14 MIR tasks; a feature extractor, not a detector
2	AI Music Arms Race [7]*	CNN/spectrogram classifiers; cross-generator protocol	Real + Suno/Udio; full-song detection	F1 0.99 $\rightarrow$ 0.629 cross-generator; no explainability
3	SONICS [14]	SpecTTTra spectro-temporal transformer	SONICS 97k songs; end-to-end SSD	+8% F1 vs. ViT on long songs; Suno/Udio only
4	Afchar et al. [2]	Neural-decoder fingerprint detector	Auto-encoded music; AIGM	99.8% acc. but fragile under shifts
5	FakeMusicCaps [6]	M5/RawNet2/ResNet; open-set attribution	FakeMusicCaps (5 TTM models)	Strong closed-set; weak open-set/unseen generator
6	SingFake [19]	AASIST/RawNet2/WavLM front-ends	SingFake (in-the-wild vocals)	Speech models degrade on singing; vocals only
7	XLS-R + SLS [20]	Multi-layer SSL feature selection	ASVspooft 2019/2021; speech	SOTA DF EER; speech, not music
8	WavLM + Attentive Fusion [10]	WavLM + multi-fusion attentive pooling	ASVspooft 2019 LA; speech	Attentive pooling beats avg-pool; speech only
9	Closing Explainability Gap [5]	Transformer + attention rollout	ASVspooft5 $\rightarrow$ FakeAVCeleb	Cross-corpus + interpretability; speech-centric
10	Fourier Explanation [1]	Frequency-domain artefact analysis	Real vs. AI music	Explains detectable spectral artefacts in AI music

with a detection head. Second, attention pooling is proven in speech but unexploited in music detection: Guo et al. and the SLS classifier of Zhang et al. show that learned, attention-weighted aggregation of self-supervised features outperforms average pooling, yet no AI-music detector has applied attention pooling over a music foundation model. Third, explainability and full-song music detection do not yet meet: every explainability method surveyed targets speech anti-spoofing, and the one explainable-music benchmark [12] is explicitly early-stage. The proposed system is positioned precisely at the intersection these three observations leave empty.

4. Identified Research Gaps

Synthesising the comparative analysis, four research gaps emerge that the proposed work targets directly:

- **Cross-generator generalisation gap.** Detectors collapse from F1 ~ 0.99 to ~ 0.629 when tested on an unseen generator [7, 6]. A music-foundation-model backbone with attention pooling, evaluated explicitly across SONICS (in-distribution) and FakeMusicCaps (cross-generator), offers a principled route to narrowing this gap.
- **Robustness gap.** Accuracy degrades under routine transformations such as MP3 compression and pitch shifting [2, 15]. Robustness must be measured and improved rather than assumed, using degraded-audio evaluation as a first-class part of the protocol.
- **Explainability gap.** No existing AI-music detector explains why it flags a track, and the explainability literature is almost entirely speech-focused [5, 9, 17]. A three-level XAI layer – attention weights, Grad-CAM spectrograms, and SHAP layer analysis – applied to full-length music would be a substantive contribution.
- **Calibration and trust gap.** The most damaging deployment failure is a confident, wrong accusation against a human artist. Few detectors report calibration, yet calibrated confidence (for example, Expected Calibration Error) is essential to set conservative thresholds that prioritise avoiding false positives.

5. Motivation for Selecting the Base Paper

The AI Music Arms Race [7], published in TISMIR, was selected as the base paper for four reasons. *Direct problem alignment*: it is one of the few works to study full-song AI-music detection under an explicit cross-generator protocol, which is exactly the failure mode this project addresses. *Quantified, citable target*: it reports the in-distribution-to-cross-generator collapse (F1 0.99 to 0.629), giving an unambiguous baseline to improve upon and a metric against which progress can be measured. *Recency and venue*: it appears in 2025 in a reputable, peer-reviewed MIR venue and is open access, satisfying the requirement for current, high-quality sources. *A clear extension surface*: the base paper deliberately leaves robustness and explainability as open problems, which the proposed MERT-plus-attention-pooling detector and three-level XAI layer are designed to fill.

In short, the base paper defines the gap and the proposed system is engineered to close it – replacing bespoke spectrogram classifiers with a music foundation model, replacing average pooling with attention pooling, and replacing opaque scores with auditable, calibrated explanations. The surrounding literature reviewed above supplies the validated components (self-supervised front-ends, attentive pooling, SHAP/Grad-CAM/attention explanations, cross-generator evaluation) that make this extension both novel and feasible.

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